Abhay

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SUMMARY

Electrical and Computer Engineering student with strong interest in developing scalable applications, predictive models and intelligent solutions using modern technologies. Proficient in **Python and Data Analysis**, with hands-on experience in real-world projects including forecasting systems.

EDUCATION

KIET Group of Institutions **2024 – 2028**

B.Tech - Electrical and Computer Engineering - ***CGPA - 6.99***

PROJECTS

EMS and Islanding Detection | MATLAB, Jupyter Notebook, Matplotlib **2026**

- Built an EMS dashboard tracking parameters (kWh, solar SOC, vibration) for campus energy optimization.
- Implemented IEEE 1547 islanding detection using voltage, frequency and power thresholds with seamless grid-to-island transitions.
- Designed rule based EMS for savings via peak shaving and battery arbitrage.

Swift Notes | JavaScript, HTML, CSS, Python, OpenAI API **2026**

- Developed an AI-driven iOS video learning app with PencilKit notebook and Gemma 2B for auto extracting notes, formulas and cheatsheets.
- Built a FastAPI backend with Whisper transcription, RAG-based embeddings and privacy focused video processing.
- Implemented community search, trending feeds and a premium glassmorphism UI with haptic feedback.

EV Charging Demand Response | Jupyter Notebook, Pandas, NumPy, XGBoost, Matplotlib **2026**

- Developed a machine learning model for forecasting EV charging demand using time series data analysis.
- Performed data preprocessing, feature engineering and model optimization to improve prediction accuracy.
- Implemented XGBoost and statistical models to identify demand patterns and trends.
- Visualized insights using data visualization tools to support infrastructure planning and decision making.

Electricity Load Forecasting | Jupyter Notebook, Pandas, NumPy, Matplotlib **2025**

- Designed a time series forecasting model to predict electricity demand using historical consumption data.
- Applied data preprocessing techniques and feature engineering to enhance model performance.
- Analyzed load patterns and trends using machine learning techniques.
- Developed visualizations to support energy planning and operational decision making.

ACHIEVEMENTS

- Research Paper accepted for publication at **2026 3rd International Conference on Advancements and Key Challenges in Green Energy and Computing (AKGEC2026)**.
- Finalist in **Developers Night Hackathon** organized by KIET. .
- Solved **150+** Data Structures and Algorithms (DSA) problems across platforms like LeetCode, GeeksforGeeks and HackerRank.

TECHNICAL SKILLS

Languages: Python, C, C++
Technologies: MATLAB Simulink, Microcontroller, Power BI
Machine Learning: LSTM, XGBoost, Pandas, NumPy, Data Analysis, Feature Engineering, Model Optimization
Developer Tools: Git, GitHub, VS Code, PyCharm, Jupyter Notebook

CERTIFICATIONS

- NPTEL Online Certification : Completed 8-week course in programming in C++.
- Simulink Onramp – MATLAB (MathWorks Certified) fundamental training in system modeling and simulation.
- GenAI-Powered Data Analytics Job Simulation - Forage.